Exploring Human Steering Performance on Complex 3D Paths

A Requirements Analysis for a Human-Centered Evaluation System

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CSE 6275 / CSE 6391: AHCI — Assignment 1

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Introduction

Background & Motivation

- **Steering Law** extends Fitts' Law to trajectory-based tasks:
  $MT = a + b \cdot \frac{L}{W}$
- Modern systems increasingly involve complex 3D environments:
  VR, simulation training, 3D design, robotic teleoperation
- Existing models focus on 2D or simple straight 3D paths
- Path *curvature* and *complexity* significantly affect performance
- **Human-Centered Machine Learning (HCML)** aims to model and support human behaviour while preserving transparency and user control

Objectives & Contributions

Objectives

1. Identify user needs for a 3D steering evaluation system
2. Develop data-driven personas and scenarios
3. Specify requirements balancing rigour, data quality & usability
4. Establish traceability between goals, needs & system capabilities

Key Contributions

- Empirical understanding of 3D steering challenges
- Personas & scenarios grounded in human-centred design
- Structured HCML requirements specification
- Usability & UX goals for high-quality data collection

Related Work

Related Work & Theoretical Background

HCI Foundations

- Interaction design: personas, scenarios, task analysis
- Goal-directed design for usable systems

Steering Law

- Accot & Zhai (1997): trajectory-based movement model
- Extensions for curvature, path complexity
- Limited 3D validation

HCML & Trajectory Prediction

- Deep learning for human trajectory prediction
- ML for VR and teleoperation behaviour modelling
- Focus on transparency, interpretability & human oversight

Methodology

Methodology: Data Gathering

Data Gathering Methods

- Systematic literature review
- Secondary source analysis
- AI-supported knowledge synthesis
- Researcher reflexive analysis

Ethical Considerations

- Informed consent
- Anonymity & privacy
- Secure data protection
- Participant well-being
- AI-specific ethical safeguards

Stakeholder Identification

Primary

HCI Researchers · Study Participants · System Operators

Secondary

VR / 3D Interface Designers · Academic Researchers

Indirect

End-users of future 3D systems (VR, simulation, training)

Porting from 2D to 3D

The Formula for porting into 3D

$$x^2 + y^2 = r^2$$

$$z = \sum_{i=1}^{n} \sin(a_i x + b_i y)$$

Comparison of 2D and d approaches

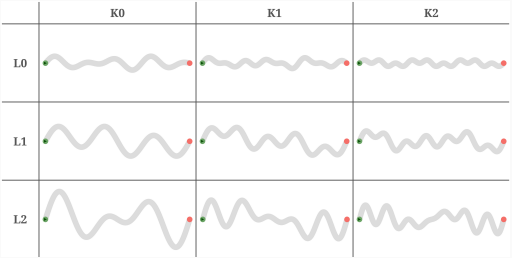


Figure 1: 2D Path Diagram (as shown by Chen et.al.)

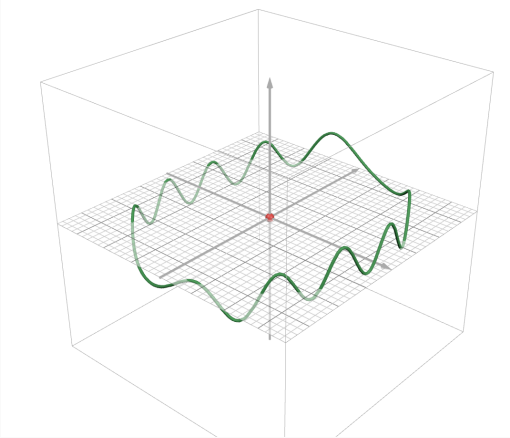


Figure 2: 3D Conversion of the 2D path Diagram

Key Findings

Key Findings

User Goals & Motivations

- Reliable, reproducible performance measures
- Clear task structure & meaningful feedback
- Intuitive interaction, low cognitive overhead

Pain Points

- No standardised 3D path representations
- Hard to compare across input devices
- Limited 3D trajectory visualisation

AI Attitudes

- Valuable for pattern discovery
- Concerns about opaque automation
- Transparency & explainability critical

Contextual Constraints

- Lab vs. non-lab contexts
- Hardware heterogeneity
- Recruitment & scheduling

Personas

Persona 1: Experimental HCI Researcher (Primary)

Dr. Nayeem Rahman, 29, PhD Candidate

Skill: High (Unity, statistics, experimental design)

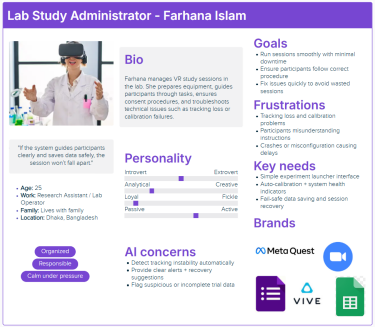
Context: University lab experiments

Goals: Reliable & reproducible data; compare path types

Pain Points: Participant confusion; manual log cleaning; difficult reproducibility

AI Expectations: Automated outlier detection; transparent analysis

Persona 2: Lab Study Administrator (Support)



Farhana Islam, 25, Research Assistant

Skill: Medium (VR setup, not advanced programming)

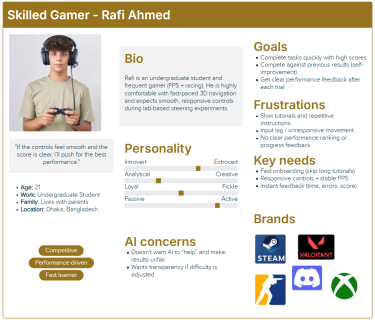
Context: Physical lab, managing participants & hardware

Goals: Smooth sessions; correct procedures; quick issue resolution

Pain Points: Tracking loss; participant confusion; crashes

AI Expectations: Automatic instability detection; clear recovery alerts

Persona 3: Skilled Gamer Participant (Secondary)



Rafi Ahmed, 21, Undergraduate (frequent gamer)

Skill: Medium-High

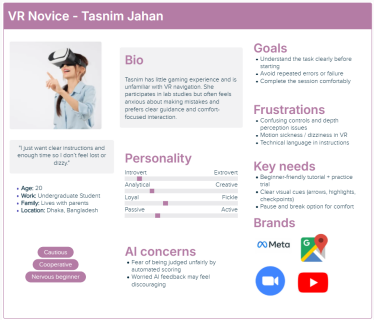
Context: Lab study with competitive mindset

Goals: Perform well; clear scoring; feel challenged

Pain Points: Long tutorials; input lag; no performance comparison

AI Concerns: Unfair difficulty adjustment

Persona 4: VR Novice Participant (Secondary)



Tasnim Jahan, 20, Undergraduate (non-gamer)

Skill: Low-Medium

Context: Lab study, unfamiliar with VR

Goals: Clear instructions; avoid failure; comfortable session

Pain Points: Confusing controls; motion sickness; technical jargon

AI Concerns: Fear of being judged by AI scoring

Persona 5: Accessibility-Constrained (Edge)

Mahmudul Hasan, 26, Graduate Student

Skill: Medium **Constraint:** Mild hand tremor

Context: Lab study with fatigue sensitivity

Goals: Participate without discomfort; fair evaluation

Pain Points: Tremor interpreted as errors; no pause/accessibility support

AI Concerns: Outlier detection may unfairly discard trials

Scenarios & Task Analysis

Usage Scenarios

#	Persona	Scenario Highlights
1	Dr. Nayeem	Configure experiment, pilot study, HCML-assisted curvature–error analysis
2	Farhana	Checklist launch, tracking-loss auto-pause, session report
3	Rafi	Quick onboarding, speed–accuracy trade-off, real-time feedback
4	Tasnim	Guided tutorial, break support, simple performance explanation
5	Mahmudul	Accessibility mode, widened tolerance, transparent outlier review

Each scenario illustrates **normal use**, **ML-assisted insight**, **accessibility**, or **operational workflows** — ensuring ML supports, not replaces, human decision-making.

Hierarchical Task Analysis

Primary Task: *Complete a 3D Steering Trial*

1. **Receive Instructions**
   Read description, learn controls
2. **Complete Practice Trial**
   Navigate practice path, get feedback
3. **Execute Experimental Trial**
   Position $\rightarrow$ steer $\rightarrow$ reach end point
4. **Receive Feedback**
   Metrics, comparison, prepare next

HCML Support Opportunities

- ML modelling of movement time & deviation
- Detection of low-quality trials
- Steering strategy clustering
- Interpretable visual summaries

Risks

- Model bias with small datasets
- Correlation causation
- Over-reliance on automation

Requirements

Functional Requirements (FR1–FR12)

FR1 Built in Unity engine

FR2 Configurable 3D path geometry

FR3 Mouse & VR controller input

FR4 Millisecond-precision timing

FR5 Boundary violation detection

FR6 High-rate trajectory capture

FR7 Tutorial & practice trials

FR8 Automatic trial randomisation

FR9 CSV / JSON data export

FR10 Descriptive stats & ML-ready data

FR11 Meta VR headset support

FR12 Standard mouse input support

Non-Functional & Other Requirements

Non-Functional (NFR1–NFR8)

- 60 FPS on GTX 1650
- Sub-perceptual input latency
- Windows 10+; Meta VR headsets
- Tutorial 3 min; data loss ¡ 0.1%
- Full logging for reproducibility
- Secure & anonymised storage

User (UR1–UR4)

- Visual path following on screen/VR
- Mouse or VR controller capable
- Researcher: experimental design knowledge
- Operator: basic Unity familiarity

Data & Environmental (DER1–DER4)

- 60 Hz sampling; Quest headsets
- Min spec: i5, 8 GB, GTX 1650
- 100 MB storage per participant

Usability & UX Goals

Usability & User Experience Goals

Usability Goals

- **Effectiveness:** 95% trial completion without help
- **Efficiency:** Setup ¡ 5 min; experiment config ¡ 10 min
- **Learnability:** Stable performance within 3 practice trials
- **Error Tolerance:** Auto-recovery; zero data loss from SW failure
- **Data Integrity:** Synchronised timestamps; verified logs

Experiential Goals

- **Trust:** 85% agreement on fair measurement
- **Confidence:** Transparent logs & visualisations
- **Control:** Latency ¡ 50 ms; 60 FPS
- **ML Transparency:** Explainable automated analysis
- **Engagement:** Breaks, progress indicators
- **Ethics:** Full informed consent; no PII stored

Traceability & Volere Shells

Requirements–Goals Traceability Matrix

User Goal	Persona / Scenario	Req. ID	UX Goal
Collect reliable data	P1 – Researcher (S1)	FR4,5,6; NFR6	Effectiveness, Trust
Configure experiments	P1 – Researcher (S1)	FR1,2,8; UR3	Efficiency, Learnability
Run sessions smoothly	P2 – Operator (S2)	FR7,9; NFR7; DER2	Efficiency, Confidence
Prevent data loss	P2 – Operator (S2)	NFR7,8; FR9	Trust, Reliability
Understand tasks clearly	P4 – Novice (S4)	FR7; UR1; NFR5	Learnability, Comfort
Complete trials quickly	P3 – Gamer (S3)	NFR1,2; FR3	Efficiency, Engagement
Participate w/o discomfort	P5 – Access. (S5)	UR1; DER1; NFR5	Comfort, Control
Ensure fair evaluation	P5 – Access. (S5)	NFR6; FR10	Trust, Transparency

Volere Requirement Shells (Examples)

FR5 – Boundary Violation Detection

- **Type:** Functional
- **Description:** Detect & count boundary violations during steering
- **Rationale:** Key performance metric for steering law experiments
- **Fit Criterion:** 99% detection accuracy
- **Priority:** High
- **Dependencies:** FR2, DER1

NFR1 – Performance (FPS)

- **Type:** Non-Functional
- **Description:** Maintain 60 FPS on GTX 1650
- **Rationale:** Stable FPS ensures accurate measurement & avoids latency bias
- **Fit Criterion:** FPS 60 in 95% of trials
- **Priority:** High
- **Dependencies:** FR2, FR3

Discussion & Conclusion

Design Trade-offs & Implications

- **Model Adaptivity vs. Experimental Control**
  Adaptive ML may introduce bias — must be optional & fully logged
- **Data Richness vs. System Performance**
  High-frequency logging improves ML but increases compute load
- **Automation vs. Human Oversight**
  Researchers must remain in control of model selection & interpretation
- **Standardisation vs. Model Generalisation**
  Standard paths aid reproducibility; diverse paths aid ML generalisation

System will NOT support: Fully autonomous design; real-time protocol changes; black-box models without interpretability.

Limitations & Future Work

Limitations

- Personas from literature, not large-scale empirical data
- ML requirements may evolve post-prototype
- Hardware scope limited to mid-range & Meta VR
- Limited cross-cultural analysis

Future Work

- Unity-based prototype for data collection
- ML models for movement time & error prediction
- Interpretable steering behaviour models
- Pilot user studies to validate requirements
- Adaptive feedback guided by HCML principles

Conclusion

- Presented a **structured requirements analysis** for an HCML system to study human steering in complex 3D environments
- Developed **5 personas** and **5 aligned scenarios** grounded in human-centred design
- Specified **functional, non-functional, user, data & environmental** requirements with traceability
- Emphasised **transparency, interpretability, human oversight** and **ethical data handling**
- Provides a foundation for **predictive modelling** of 3D steering performance and **adaptive, human-centred systems**

Thank You!

Questions?

References